

# LINKEDLIST

Q7)

## Question:

| DS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Session | Linked List | Question Information | Level 1 | Challenge 37 |
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| <p><b>Question description</b></p> <p>the popular engineering college got lowest pass percentage in last semester. the principal conducted faculty meeting and decided to visit all the classes surprisingly.</p> <p>Dr.Ramprasath is a faculty, who handling data structure course for EEE department second year students.</p> <p>one day this faculty was handling very interesting topic in data structure such that Linked List,</p> <p>During this lecture time, principal surprisingly visited to the class and asking to conduct surprise test on Linked list concept.</p> <p>So the faculty decided to conduct test on the topic of Linked List.</p> <p>the question was given to last bench students that is,</p> <p>The nodes are deleted before a certain given node in the linked list.</p> <p>For example if the given Linked List is 5-&gt;10-&gt;15-&gt;20-&gt;25 and</p> <p>delete before 15 then the Linked List becomes 15-&gt;20-&gt;25.</p> <p><b>Constraint :</b></p> <p><math>1 &lt; N &lt; 1000</math></p> <p><math>1 &lt; P &lt; N-1</math></p> <p><b>INPUT Format</b></p> <p>First line contains the number of datas- N.</p> <p>Second line contains N integers(the given linked list).</p> |         |             |                      |         |              |

## PROGRAM:

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  struct Node {
5      int data;
6      struct Node *next;
7  };
8
9  struct Node* createNode(int data) {
10     struct Node *newNode = (struct Node*)malloc(sizeof(struct Node));
11     newNode->data = data;
12     newNode->next = NULL;
13     return newNode;
14 }
15
16 struct Node* insertEnd(struct Node *head, int data) {
17     struct Node *newNode = createNode(data);
18
19     if (head == NULL)
20         return newNode;
21
22     struct Node *temp = head;
23
24     while (temp->next != NULL)
25         temp = temp->next;
26
27     temp->next = newNode;
28
29     return head;
30 }
31
32 struct Node* deleteAtPosition(struct Node *head, int position, int *valid) {
33     if (head == NULL || position < 0) {
34         *valid = 0;
35         return head;
36     }
37
38     if (position == 0) {
39         struct Node *temp = head;
40         head = head->next;
41         free(temp);
42         return head;
43     }
44
45     struct Node *temp = head;
46
47     for (int i = 0; i < position - 1; i++) {
48         if (temp == NULL || temp->next == NULL) {
49             *valid = 0;
50             return head;
51         }
52
53         temp = temp->next;
54     }
55
56     if (temp->next == NULL) {
57         *valid = 0;
58         return head;
59     }
60 }
```

```

61     struct Node *deleteNode = temp->next;
62     temp->next = deleteNode->next;
63     free(deleteNode);
64
65     return head;
66 }
67
68 void display(struct Node *head) {
69     printf("Linked List:");
70
71     while (head != NULL) {
72         printf("->%d", head->data);
73         head = head->next;
74     }
75
76     printf("\n");
77 }
78
79 int main() {
80     int N, data, P;
81     int valid = 1;
82     struct Node *head = NULL;
83
84     scanf("%d", &N);
85
86     for (int i = 0; i < N; i++) {
87         scanf("%d", &data);
88         head = insertEnd(head, data);
89     }
90
91     scanf("%d", &P);
92
93     head = deleteAtPosition(head, P, &valid);
94
95     if (!valid)
96         printf("Invalid Node! ");
97
98     display(head);
99
100     return 0;
101 }

```

## OUTPUT:

### TEST CASE 1:

```
4
9 77 12 6
1
Linked List:->9->12->6
```

### TEST CASE 2:

```
4
91 77 12 61
2
Linked List:->91->77->61
```